

SET 28 – Vehicle Performance Analysis

Q1. Violin plot of Fuel Efficiency by Safety Rating

```
# Install package if required
```

```
# install.packages("ggplot2")
```

```
library(ggplot2)
```

```
data <- data.frame(
```

```
  Vehicle_ID = c("V1","V2","V3","V4","V5"),
```

```
  Engine_Size = c(1.5,2.0,3.0,2.5,1.8),
```

```
  Horsepower = c(110,150,250,200,130),
```

```
  Fuel_Efficiency = c(18,15,12,14,17),
```

```
  Top_Speed = c(180,200,250,220,190),
```

```
  Safety_Rating = c(4,5,5,4,3)
```

```
)
```

```
ggplot(data, aes(x=factor(Safety_Rating),
```

```
  y=Fuel_Efficiency,
```

```
  fill=factor(Safety_Rating))) +
```

```
geom_violin(trim=FALSE) +
```

```
geom_boxplot(width=0.15, fill="white") +
```

```
labs(
```

```
  title="Fuel Efficiency Distribution by Safety Rating",
```

```
  x="Safety Rating",
```

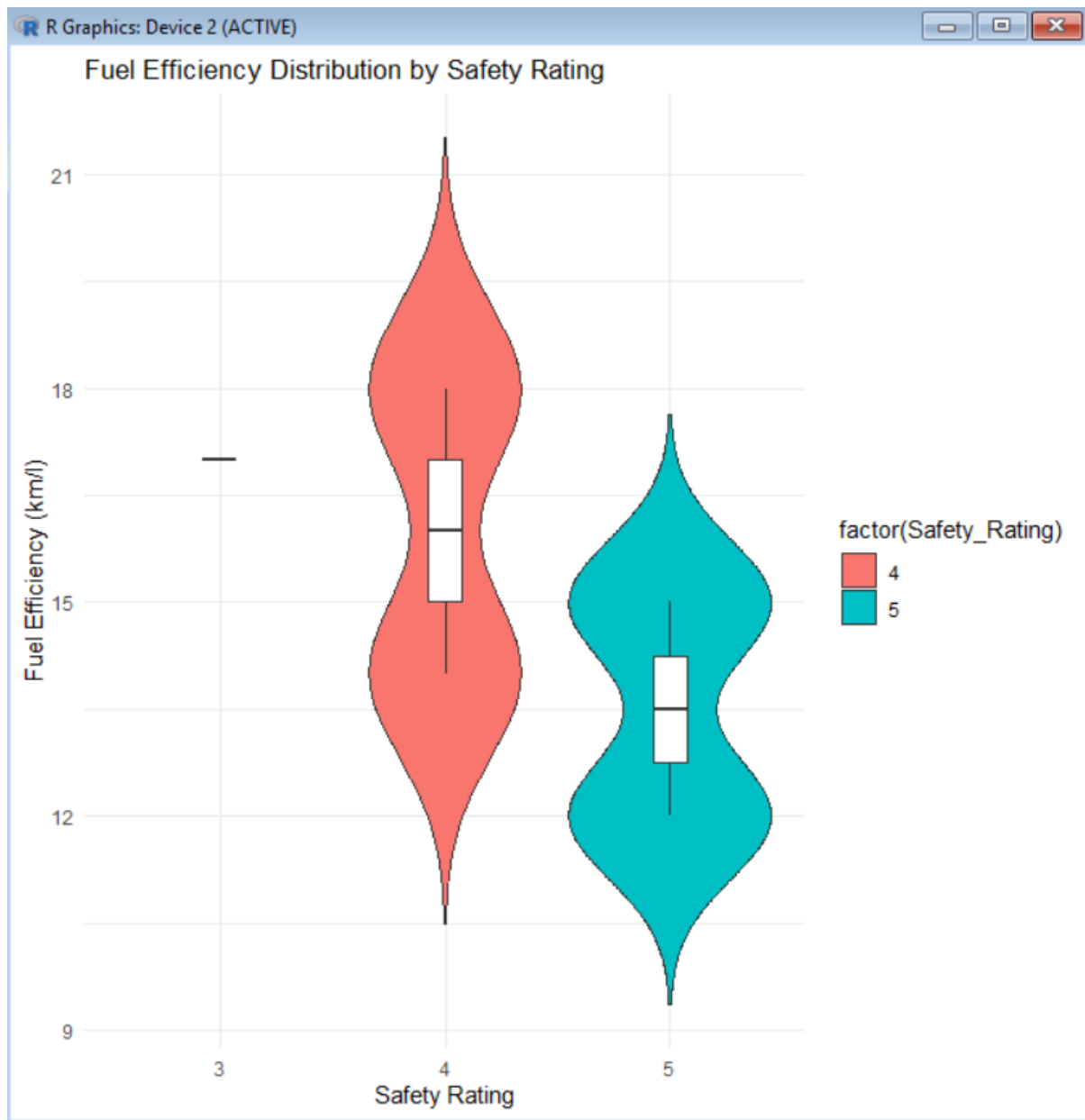
```
  y="Fuel Efficiency (km/l)"
```

) +

theme_minimal()

Interpretation: Vehicles with lower safety ratings in this small dataset tend to have higher fuel efficiency, while vehicles with higher safety ratings show lower fuel efficiency.

OUTPUT;



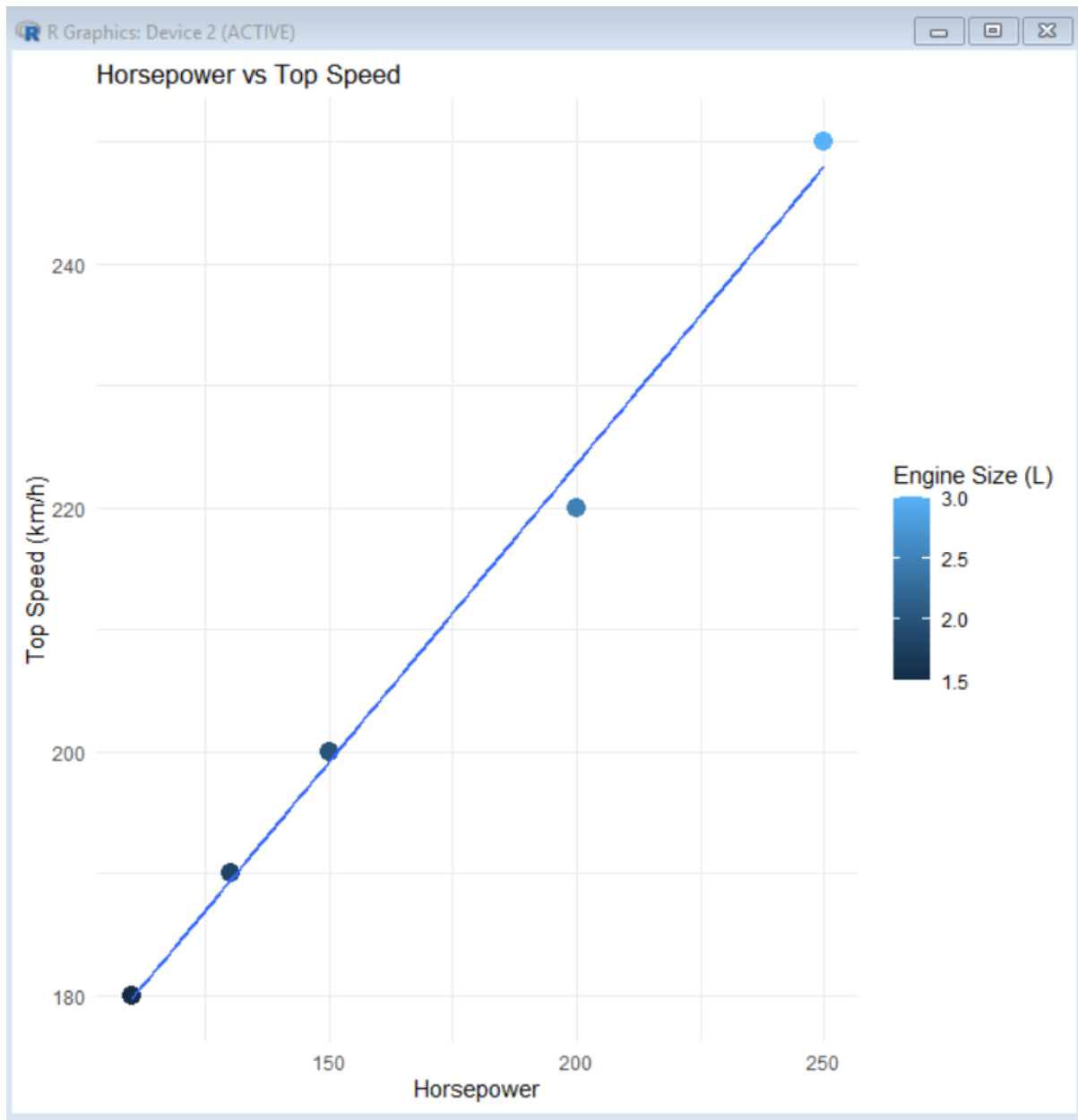
Q2. Horsepower vs Top Speed

library(ggplot2)

```
ggplot(data, aes(x=Horsepower,  
                y=Top_Speed,  
                color=Engine_Size)) +  
  geom_point(size=4) +  
  geom_smooth(method="lm", se=FALSE) +  
  labs(  
    title="Horsepower vs Top Speed",  
    x="Horsepower",  
    y="Top Speed (km/h)",  
    color="Engine Size (L)"  
  ) +  
  theme_minimal()
```

Interpretation: The plot shows a strong positive relationship. As horsepower increases, top speed generally increases.

OUTPUT;



Q3. Correlation Heatmap

```
library(ggplot2)
```

```
library(reshape2)
```

```
num_data <- data[, c(  
  "Engine_Size",  
  "Horsepower",  
  "Fuel_Efficiency",
```

```
"Top_Speed",  
"Safety_Rating"  
)]
```

```
cor_matrix <- cor(num_data)
```

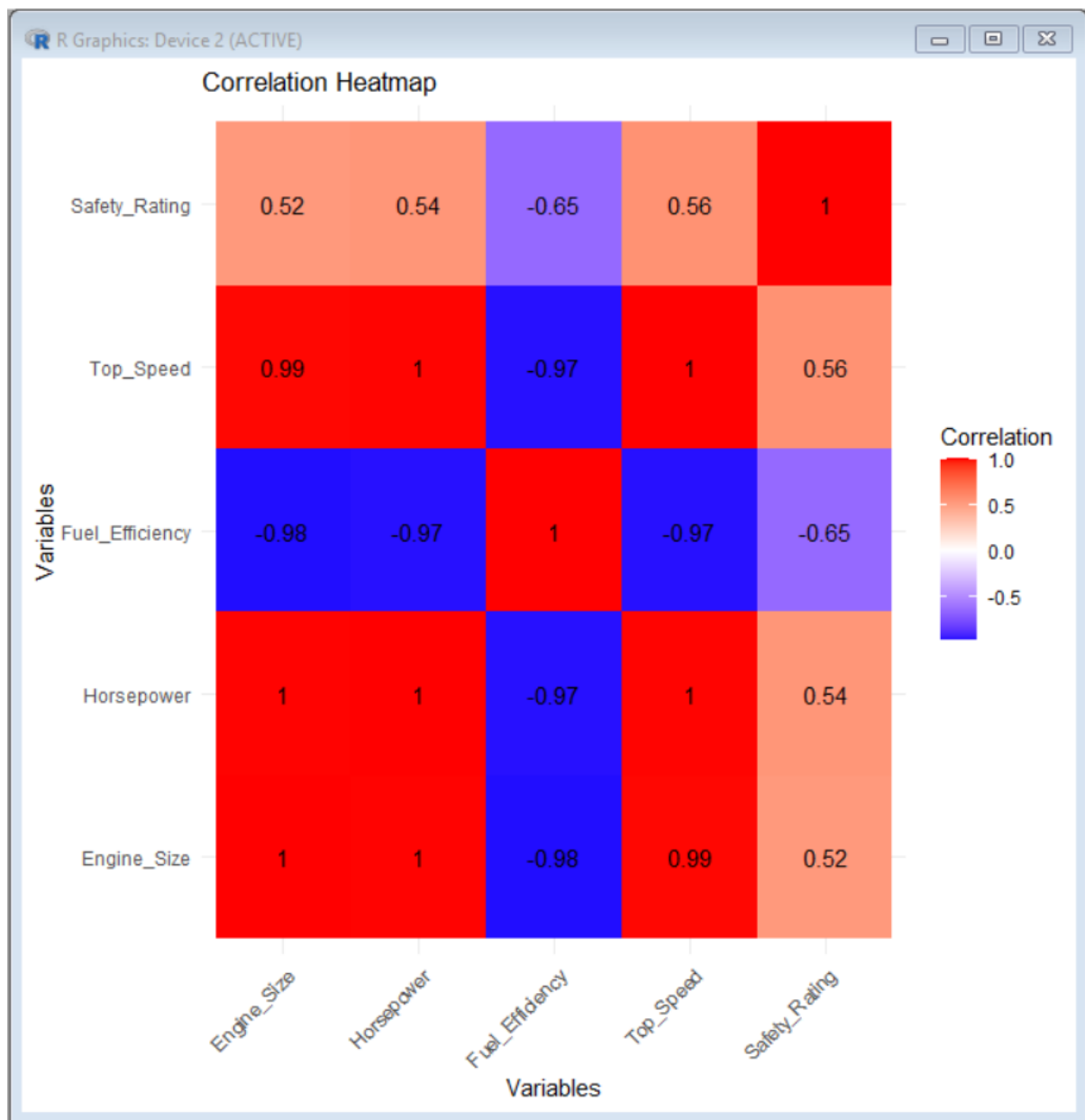
```
cor_data <- melt(cor_matrix)
```

```
ggplot(cor_data,  
  aes(x=Var1, y=Var2, fill=value)) +  
  geom_tile() +  
  geom_text(aes(label=round(value,2)), color="black") +  
  scale_fill_gradient2(  
    low="blue",  
    mid="white",  
    high="red",  
    midpoint=0  
  ) +  
  labs(  
    title="Correlation Heatmap of Vehicle Performance",  
    x="Variables",  
    y="Variables",  
    fill="Correlation"  
  ) +  
  theme_minimal() +
```

```
theme(axis.text.x=element_text(angle=45, hjust=1))
```

Interpretation: Horsepower and Top Speed have a strong positive correlation. Fuel Efficiency has a negative relationship with horsepower and top speed.

OUTPUT;



Q4. Tableau Dashboard

This question is for **Tableau**, so there is no R code required.

Create:

- Bar chart → Vehicle ID vs Fuel Efficiency
- Scatter plot → Horsepower vs Top Speed
- Filter → Engine Size
- Filter → Safety Rating
- Add both charts to one dashboard.

SET 29 – Student Academic Performance

Dataset from the question:

Q1. Stacked Area Chart

```
library(ggplot2)
```

```
library(reshape2)
```

```
data <- data.frame(
  Student_ID = c("S1","S2","S3","S4","S5"),
  Age = c(19,21,20,22,23),
  Study_Hours = c(12,8,15,10,7),
  Attendance = c(90,70,95,85,60),
  Test_Score = c(85,70,92,80,65),
  Participation_Score = c(8,7,9,8,6)
)
```

```
area_data <- data[, c(
  "Student_ID",
  "Test_Score",
  "Participation_Score"
```

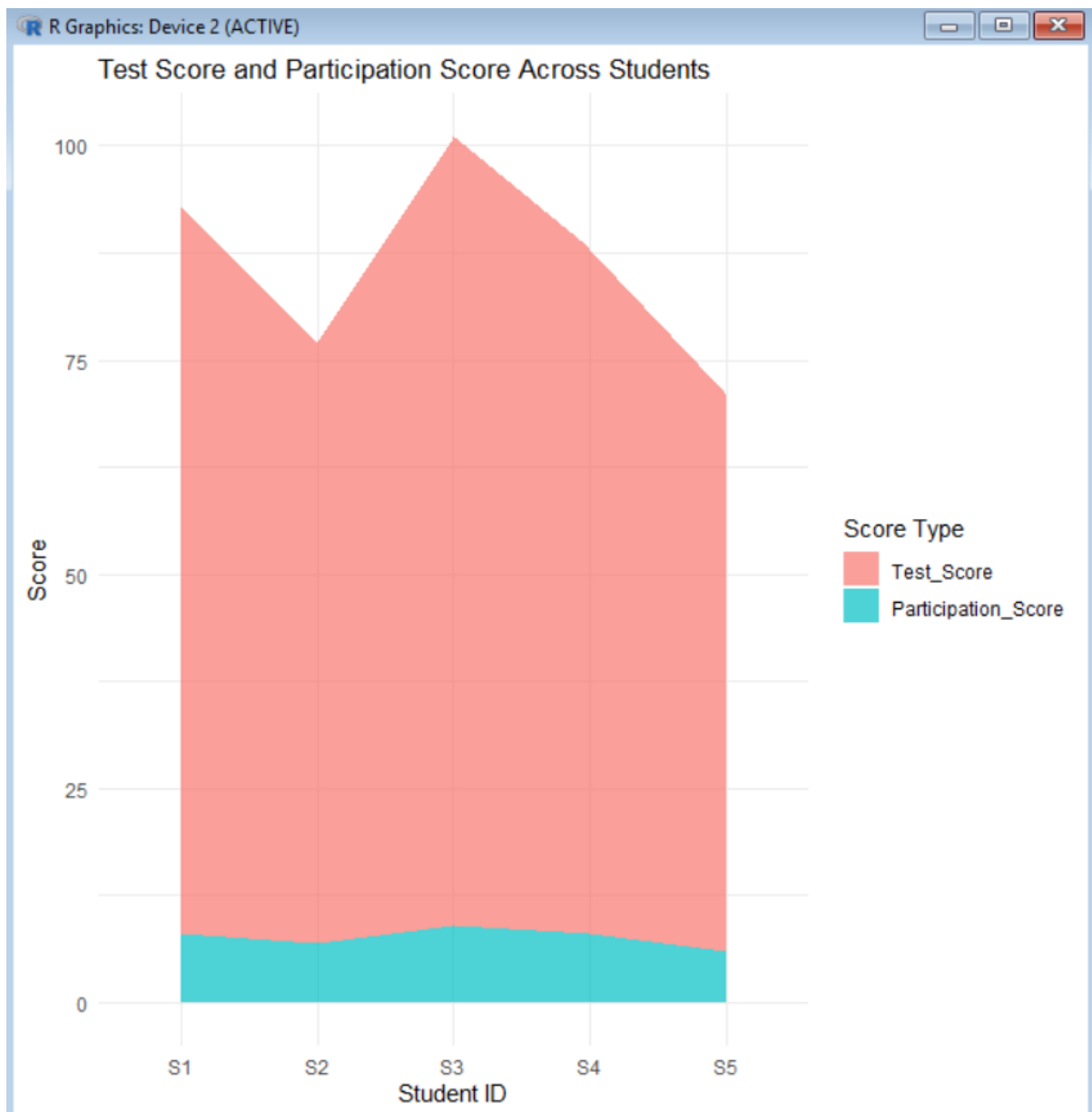
```
)]
```

```
area_long <- melt(  
  area_data,  
  id.vars="Student_ID",  
  variable.name="Score_Type",  
  value.name="Score"  
)
```

```
ggplot(area_long,  
  aes(x=Student_ID,  
      y=Score,  
      fill=Score_Type,  
      group=Score_Type)) +  
geom_area(alpha=0.7) +  
labs(  
  title="Test Score and Participation Score Across Students",  
  x="Student",  
  y="Score",  
  fill="Score Type"  
) +  
theme_minimal()
```

Interpretation: Students S3 and S1 show relatively strong academic performance. S5 has the lowest test and participation scores.

OUTPUT;



Q2. Boxplot of Study Hours by Attendance Quartiles

```
library(ggplot2)
```

```
data$Attendance_Quartile <- cut(  
  data$Attendance,  
  breaks=quantile(  
    data$Attendance,  
    probs=c(0,0.25,0.5,0.75,1),
```

```

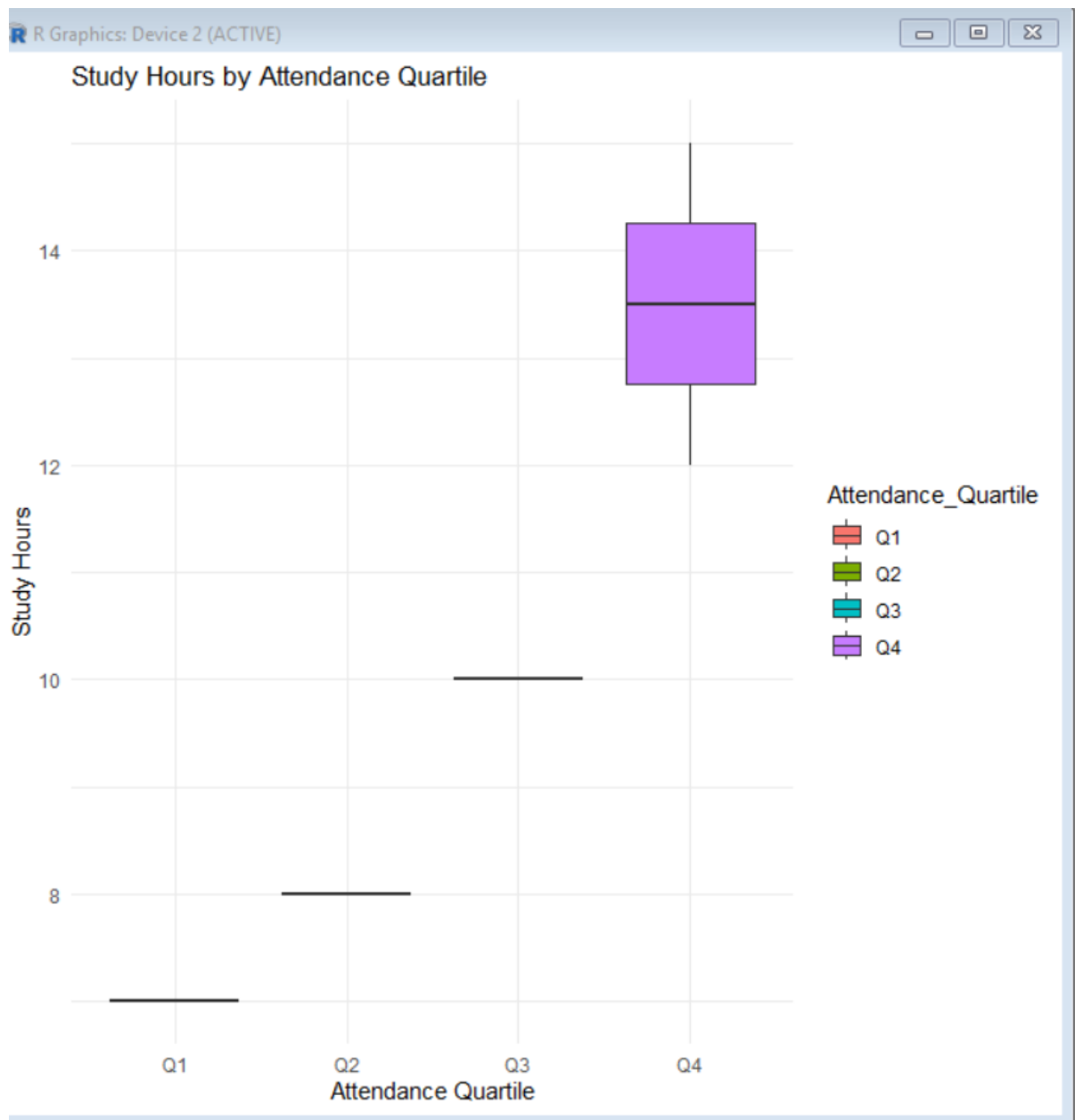
na.rm=TRUE
),
include.lowest=TRUE,
labels=c("Q1","Q2","Q3","Q4")
)

ggplot(data,
  aes(x=Attendance_Quartile,
      y=Study_Hours,
      fill=Attendance_Quartile)) +
geom_boxplot() +
labs(
  title="Study Hours by Attendance Quartile",
  x="Attendance Quartile",
  y="Study Hours"
) +
theme_minimal()

```

Interpretation: Higher attendance generally corresponds to higher study hours in this dataset. Because there are only five students, the quartile groups contain very few observations.

OUTPUT;



Q3. Density Plot for Test Score

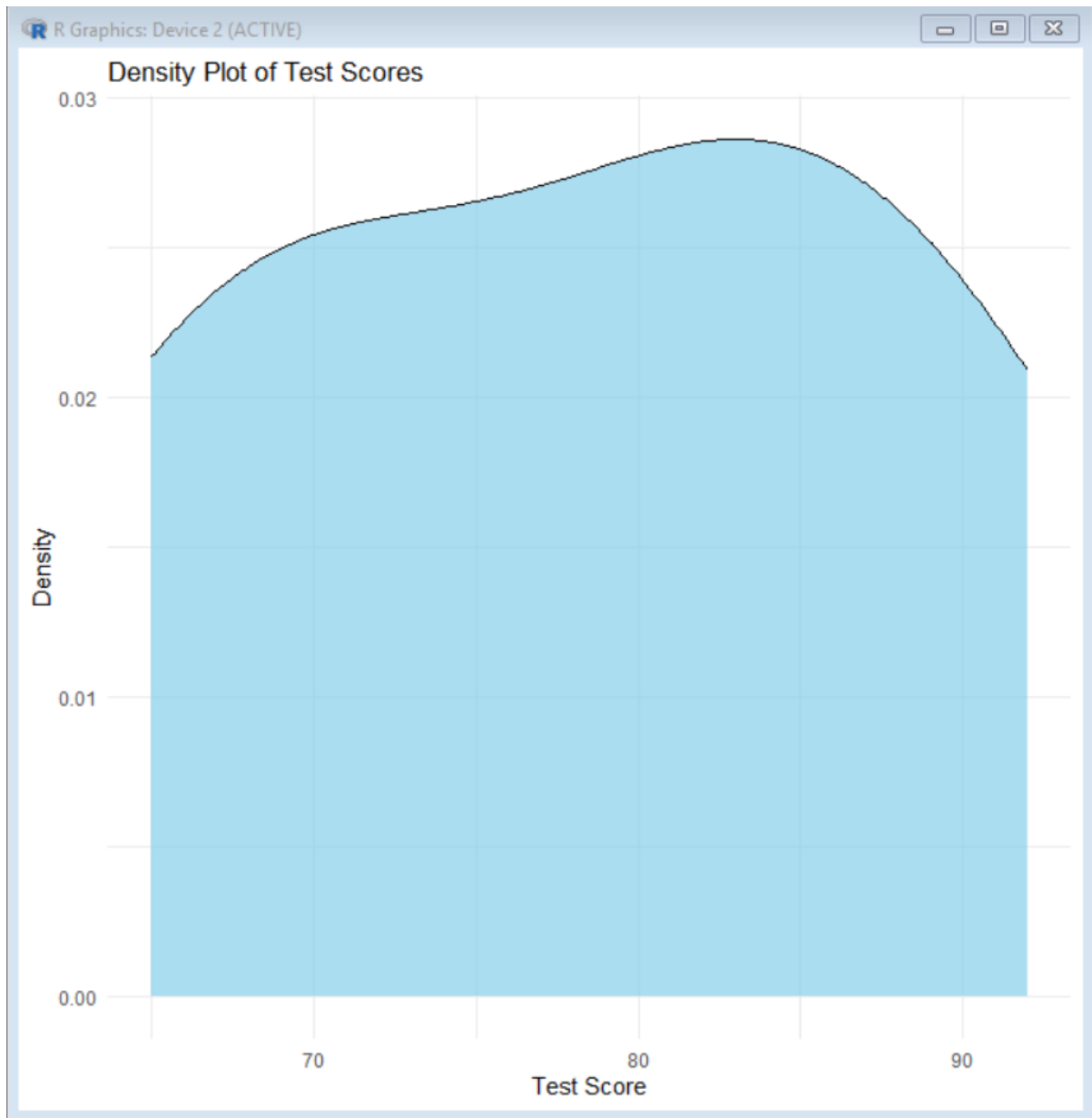
```
library(ggplot2)
```

```
ggplot(data, aes(x=Test_Score)) +  
  geom_density(  
    fill="skyblue",  
    alpha=0.6  
  ) +
```

```
labs(  
  title="Density Plot of Test Scores",  
  x="Test Score",  
  y="Density"  
) +  
theme_minimal()
```

Interpretation: The density plot shows the distribution of test scores. The small dataset makes it difficult to determine a reliable overall distribution or identify strong outliers.

OUTPUT;



Q4. Tableau Interactive Scatter Plot

This is a **Tableau question**.

Use:

- X-axis → Study_Hours
- Y-axis → Test_Score
- Color → Participation_Score
- Filter → Age
- Add the scatter plot to a dashboard.

SET 30 – Mobile App Usage

Dataset from the question:

Q1. Histogram and Density Plot for Screen Time

```
library(ggplot2)
```

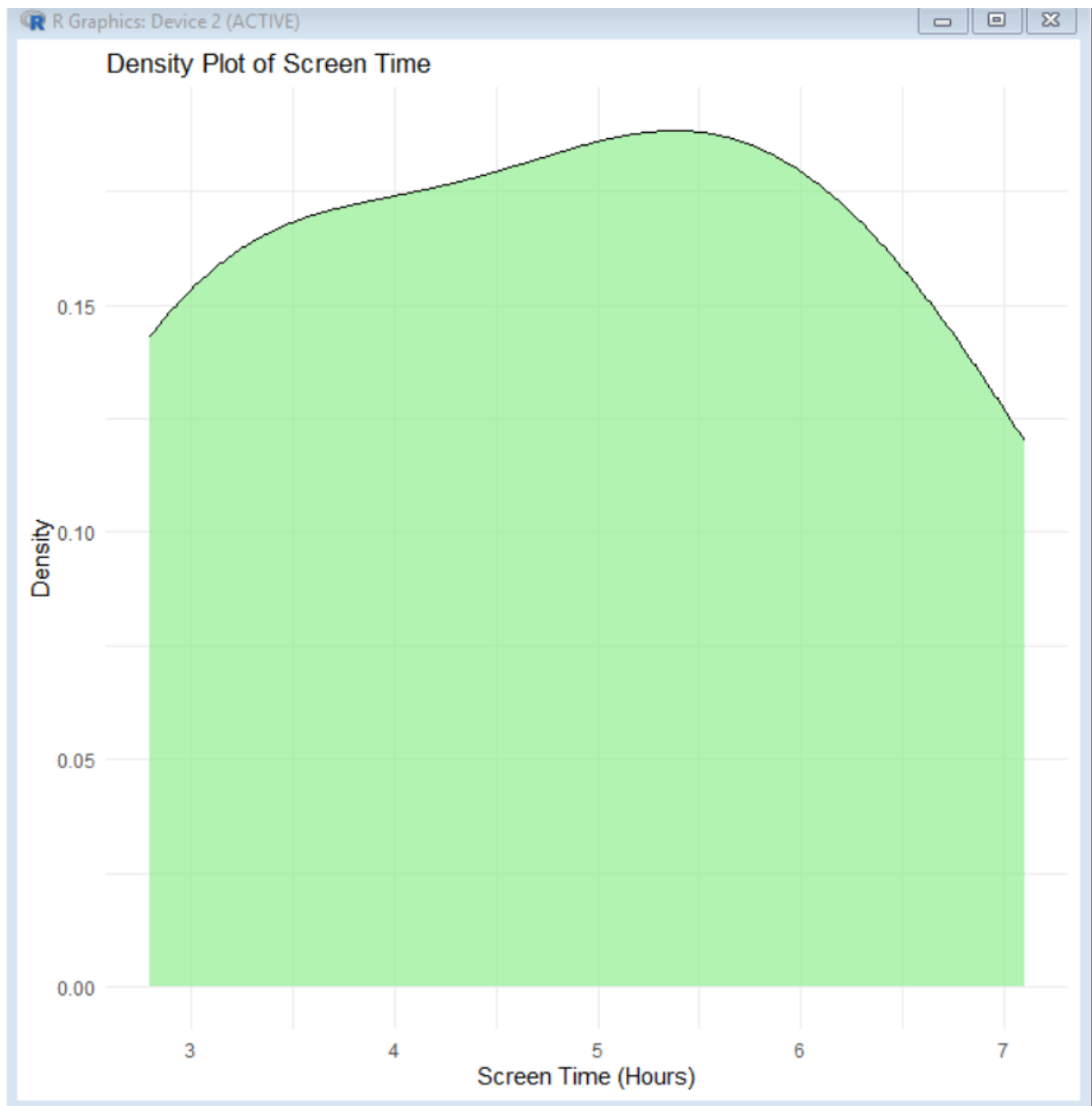
```
data <- data.frame(  
  User_ID = c("U01","U02","U03","U04","U05","U06"),  
  Gender = c("Male","Female","Male","Female","Male","Female"),  
  Age = c(20,22,19,21,23,20),  
  Screen_Time = c(4.5,6.0,3.2,7.1,2.8,5.4),  
  App_Usage_Count = c(18,25,12,30,10,22),  
  Data_Used = c(2.4,3.8,1.6,4.5,1.2,3.1),  
  Satisfaction = c(3,5,3,5,2,4),  
  Usage_Date = as.Date(c(  
    "2025-01-08",  
    "2025-01-08",  
    "2025-02-11",  
    "2025-02-11",  
    "2025-03-14",  
    "2025-03-14"  
  ))  
)
```

```
ggplot(data, aes(x=Screen_Time)) +
```

```
geom_histogram(  
  aes(y=after_stat(density)),  
  bins=6,  
  fill="steelblue",  
  color="black",  
  alpha=0.7  
) +  
geom_density(  
  color="red",  
  linewidth=1.2  
) +  
labs(  
  title="Screen Time Distribution",  
  x="Screen Time (hours)",  
  y="Density"  
) +  
theme_minimal()
```

Interpretation: Screen time ranges from approximately 2.8 to 7.1 hours. Female users in this sample generally show higher screen-time values.

OUTPUT;



Q2. Data Used vs Screen Time + Correlation

```
library(ggplot2)
```

```
correlation <- cor(  
  data$Data_Used,  
  data$Screen_Time  
)
```

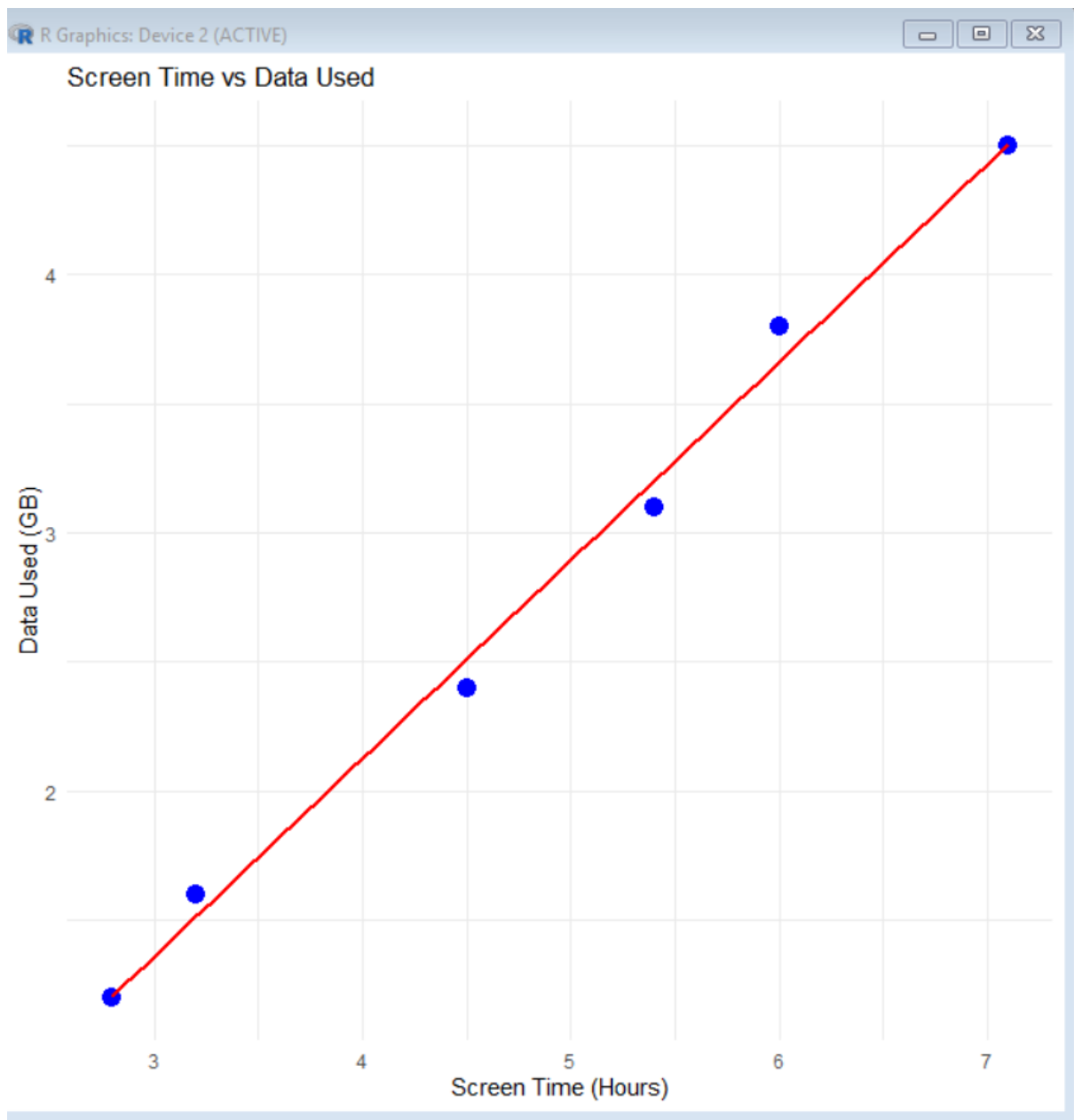


```
print(paste(
  "Correlation between Data Used and Screen Time:",
  round(correlation, 3)
))
```

```
ggplot(data,
  aes(x=Screen_Time,
    y=Data_Used)) +
  geom_point(
    size=4,
    color="blue"
  ) +
  geom_smooth(
    method="lm",
    se=FALSE,
    color="red"
  ) +
  labs(
    title="Data Used vs Screen Time",
    x="Screen Time (hours)",
    y="Data Used (GB)"
  ) +
  theme_minimal()
```

Interpretation: The relationship is strongly positive: users with greater screen time generally consume more mobile data.

OUTPUT;



Q3. Average Satisfaction by Gender

```
library(ggplot2)
```

```
library(dplyr)
```

```
avg_satisfaction <- data %>%
```

```
  group_by(Gender) %>%
```

```
  summarise(
```

```
Average_Satisfaction = mean(Satisfaction)
)

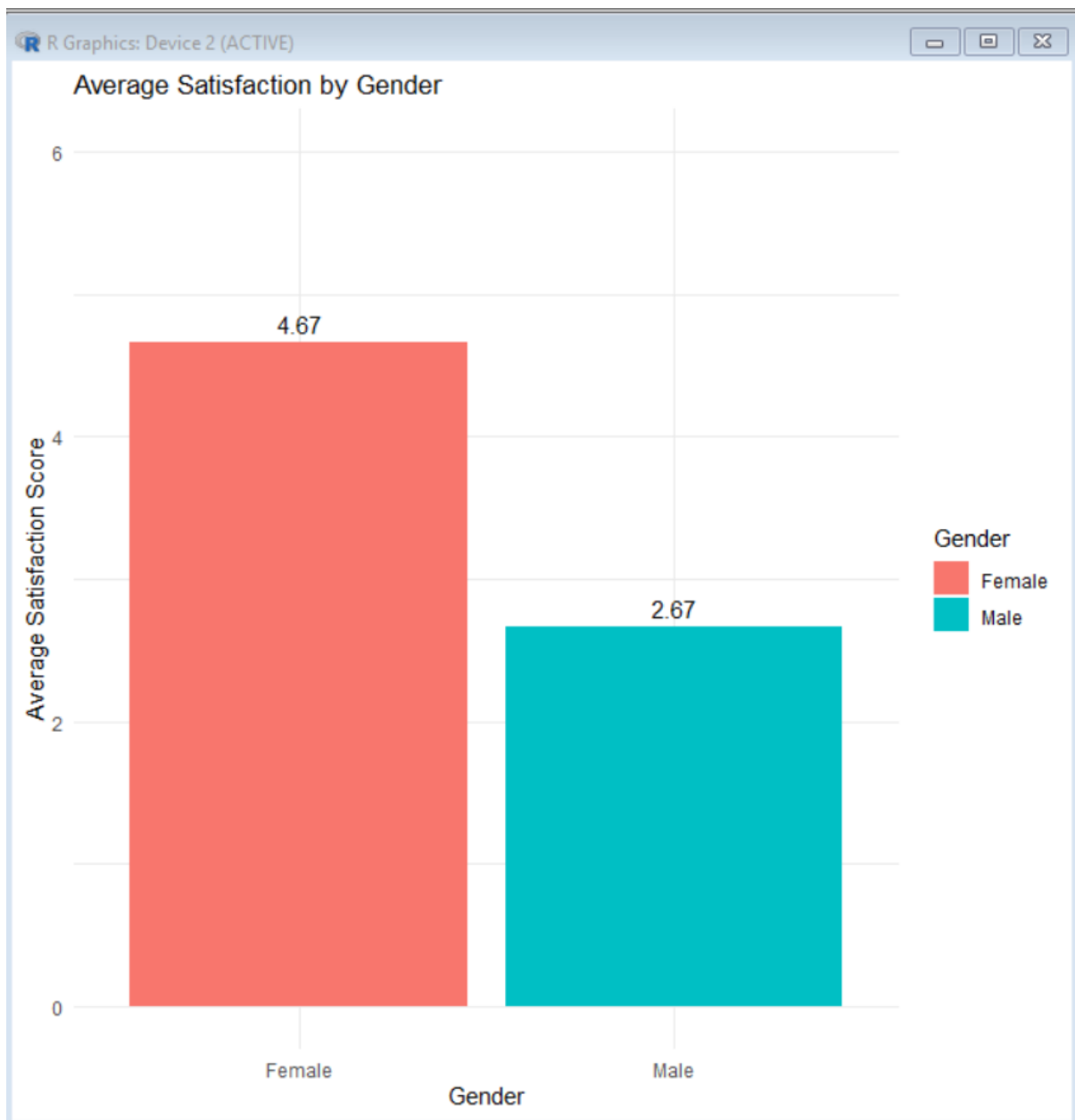
print(avg_satisfaction)

ggplot(
  avg_satisfaction,
  aes(
    x=Gender,
    y=Average_Satisfaction,
    fill=Gender
  )
) +
  geom_bar(
    stat="identity"
  ) +
  geom_text(
    aes(
      label=round(Average_Satisfaction,2)
    ),
    vjust=-0.5
  ) +
  labs(
    title="Average Satisfaction Score by Gender",
    x="Gender",
```

```
y="Average Satisfaction Score"  
) +  
ylim(0,6) +  
theme_minimal()
```

Interpretation: Female users have a higher average satisfaction score than male users in the given dataset.

OUTPUT;



Q4. Tableau Dashboard

This question is for **Tableau**.

Create the following:

Bubble Chart

- X-axis → Screen_Time
- Y-axis → Data_Used
- Color → Gender
- Size → App_Usage_Count

Line Chart

- X-axis → Usage_Date
- Y-axis → Screen_Time or Data_Used

Filters

- Gender
- Usage_Date

Then combine both charts into a **mini interactive dashboard**.